

LESSON 6-2 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Dividing a Fraction by a Fraction**

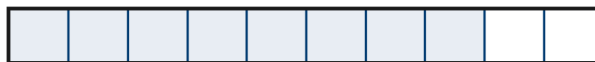
Maryland Standard 6.NS.A.1 · Learning target – I can divide a fraction by another fraction using the reciprocal.

REMEMBER

- To divide by a fraction, **multiply by its reciprocal**:
 $3/4 \div 3/8 = 3/4 \times 8/3$.
- Keep the first fraction, change $\div$ to $\times$, and flip only the **second** fraction.
- Check: quotient $\times$ divisor = dividend. $2 \times 3/8 = 6/8 = 3/4$.

WORKED EXAMPLEHow many $3/8$ -cup servings are in $3/4$ cup of yogurt?**Step 1** Write the division: $3/4 \div 3/8$ **Step 2** Multiply by the reciprocal: $3/4 \times 8/3 = 24/12$ **Step 3** Simplify: $24/12 = 2$ **Answer** 2 servings**WORD BANK****reciprocal** – the flipped fraction: $3/8$ becomes $8/3$ **quotient** – the answer when you divide**divisor** – the number you divide by**PRACTICE***show your work · label units*

1. The bar is 1 whole. The shaded part is $4/5$ of the bar. How many $1/10$ pieces fit in $4/5$?
Write the division and the quotient.

Step 1 Count the shaded tenths.**Step 2** Each shaded tenth is one $1/10$ piece.

1 whole

Division _____

2. Divide. Write each quotient in simplest form.

a. $3/4 \div 1/8 =$ _____

b. $2/3 \div 2/9 =$ _____

c. $5/6 \div 1/3 =$ _____

d. $7/10 \div 2/5 =$ _____

e. $1/2 \div 3/4 =$ _____

f. $8/9 \div 4/9 =$ _____

3. **Pancakes.** A recipe needs **$2/3$ cup** of milk. Rosa has only **$1/2$ cup**. What **fraction of the recipe** can she make?

Step 1 Write the division: $1/2 \div 2/3$.**Step 2** Flip only the second fraction, then multiply.*Show your work.***Answer** _____ of the recipe

- 4. Bookmarks.** Jae has **$\frac{5}{6}$ yard** of ribbon. Each bookmark needs **$\frac{5}{12}$ yard**. How many bookmarks can Jae make?

Step 1 Write $\frac{5}{6} \div \frac{5}{12}$.

Step 2 Multiply $\frac{5}{6}$ by $\frac{12}{5}$, then simplify.

Answer _____ bookmarks

- 5. Find the Error.** Imani wrote $\frac{4}{5} \div \frac{2}{3} = \frac{4}{5} \times \frac{2}{3} = \frac{8}{15}$. Explain her mistake. Then find the correct quotient.

Frame Imani forgot to ____ the second fraction. The correct answer is ____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME With a measuring cup, show someone at home how many $\frac{1}{4}$ -cup scoops are in $\frac{3}{4}$ cup.

LESSON 6-2 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Dividing with Mixed Numbers**

Maryland Standard 6.NS.A.1 · Learning target – I can divide mixed numbers by writing them as improper fractions first.

REMEMBER

- Write each mixed number as an **improper fraction**:
 $2\frac{3}{4} = 11/4$.
- Then multiply by the **reciprocal** of the divisor.
- Simplify. Write an improper answer as a mixed number.

WORKED EXAMPLEA board is $5\frac{1}{4}$ feet long. How many $\frac{3}{4}$ -foot pieces can be cut?**Step 1** $5\frac{1}{4} = 21/4$, so $21/4 \div 3/4 = 21/4 \times 4/3$ **Step 2** Multiply and simplify: $84/12 = 7$ **Answer 7 pieces****WORD BANK****mixed number** – a whole number and a fraction, like $2\frac{3}{4}$ **reciprocal** – the flipped fraction: $3/4$ becomes $4/3$ **improper fraction** – top number is bigger, like $11/4$ **PRACTICE***show your work · label units*

- 1.**
- Divide. Write each quotient in simplest form.

a. $4\frac{1}{2} \div 3/4 =$ _____

b. $3\frac{1}{3} \div 5/6 =$ _____

c. $2\frac{1}{4} \div 1\frac{1}{2} =$ _____

d. $6 \div 1\frac{1}{3} =$ _____

e. $5\frac{5}{6} \div 1\frac{1}{6} =$ _____

f. $1\frac{7}{8} \div 2\frac{1}{2} =$ _____

- 2.**
- Write each mixed number as an improper fraction.

Step 1 Multiply the whole number by the bottom number.**Step 2** Add the top number. Keep the bottom number.

a. $3\frac{2}{5} =$ _____

b. $4\frac{5}{6} =$ _____

- 3. Soup Bowls.**
- A pot has
- $8\frac{3}{4}$ cups**
- of soup. Each bowl holds
- $1\frac{1}{2}$ cups**
- . How many bowls can be
- filled all the way**
- ? How much soup is
- left over**
- ?

Step 1 Change both mixed numbers to improper fractions.**Step 2** Divide. Only whole bowls count.**Step 3** Subtract to find what is left.*Show your work.***Answer** _____

- 4. Hiking.** A trail is $7\frac{1}{2}$ miles long. Kenji hikes $1\frac{1}{4}$ miles each hour. How many hours will it take him to hike the whole trail?

Step 1 Write $7\frac{1}{2} = \frac{15}{2}$ and $1\frac{1}{4} = \frac{5}{4}$.

Step 2 Multiply $\frac{15}{2}$ by $\frac{4}{5}$, then simplify.

Answer _____ hours

- 5. Reason.** Without dividing, decide: is $6\frac{1}{2} \div 1\frac{1}{4}$ greater than or less than $6\frac{1}{2}$?

Frame The answer is ____ than $6\frac{1}{2}$ because $1\frac{1}{4}$ is ____ than 1.

☐ greater than $6\frac{1}{2}$

☐ less than $6\frac{1}{2}$

TEST PRACTICE

choose the best answer

- 6.** What is $3\frac{1}{2} \div 1\frac{3}{4}$?

A $\frac{1}{2}$

B 2

C $3\frac{2}{3}$

D $6\frac{1}{8}$

- 7.** A rope is $6\frac{2}{3}$ feet long. It is cut into pieces that are each $\frac{5}{6}$ foot long. How many pieces are there?

A $\frac{1}{8}$

B $5\frac{5}{9}$

C 8

D 12

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Measure $1\frac{1}{2}$ cups of water with a $\frac{1}{4}$ -cup scoop. Does the number of scoops match $1\frac{1}{2} \div \frac{1}{4}$?